SERRE'S CONDITION S_K ASCENDS ALONG A FLAT MAP WITH S_K FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Depth computed from the fibre and the base. Invoked at the start of the counted proof (lines 46633-46640). Expresses the depth of the localised ring as the sum of the fibre contribution and the depth of the base local ring.
- Dimension of the localised ring compared with the base and the fibre. Used in the final estimate (lines 46641-46673). Supplies the inequality between the dimension of the local ring and the two pieces it splits into.

Source reference. Original lemma 0337: sources/stacks-algebra/pinned-original-source.tex, source lines 46591–46591. Navigation pointer for \ref{lemma-apply-grothendieck} used inside the counted span (line 46635); the argument that consumes it is retained in the span.

Source reference. Original lemma 00ON: sources/stacks-algebra/pinned-original-source.tex, source lines 27638–27638. Navigation pointer for \ref{lemma-dimension-base-fibre-equals-total} used inside the counted span (line 46651); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00OM: sources/stacks-algebra/pinned-original-source.tex, source lines 27608–27608. Navigation pointer for \ref{lemma-dimension-base-fibre-total} used inside the counted span (line 46649); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. Assume*

(1) *R is Noetherian,*
(2) *S is Noetherian,*
(3) *φ is flat,*
(4) *the fibre rings $S \otimes_R \kappa(\mathfrak{p})$ are (S_k), and*
(5) *R has property (S_k).*

Then S has property (S_k).

Proof. Let $\mathfrak{q}$ be a prime of S lying over a prime $\mathfrak{p}$ of R. By Lemma 0337 we have

$$\operatorname{depth}(S_\mathfrak{q}) = \operatorname{depth}(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) + \operatorname{depth}(R_\mathfrak{p}).$$

On the other hand, we have

$$\dim(R_\mathfrak{p}) + \dim(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) \geq \dim(S_\mathfrak{q})$$

by Lemma 00OM. (Actually equality holds, by Lemma 00ON but strictly speaking we do not need this.) Finally, as the fibre rings of the map are assumed (S_k) we see that $\operatorname{depth}(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) \geq \min(k, \dim(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}))$. Thus the lemma follows by the following string of inequalities

$$\begin{aligned}
\operatorname{depth}(S_\mathfrak{q}) &= \operatorname{depth}(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) + \operatorname{depth}(R_\mathfrak{p}) \\
&\geq \min(k, \dim(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q})) + \min(k, \dim(R_\mathfrak{p})) \\
&= \min(2k, \dim(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) + k, k + \dim(R_\mathfrak{p}), \dim(S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}) + \dim(R_\mathfrak{p})) \\
&\geq \min(k, \dim(S_\mathfrak{q}))
\end{aligned}$$

as desired. □